Local Quadratic Convergence of Newton's Method

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1]*Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

Express the next error as an integral of derivative differences. The quadratic estimate then proves its own invariant neighborhood, avoiding a circular assumption that all iterates remain near the root. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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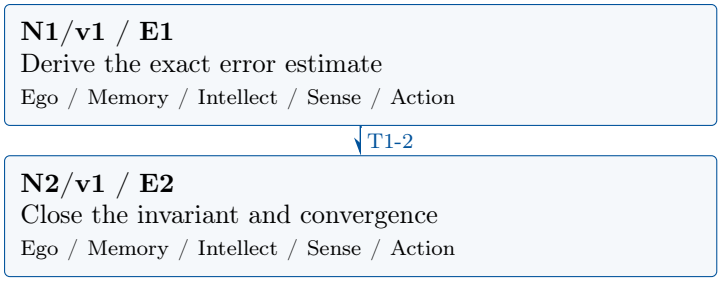
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let f be C^2 on an open interval containing $[\alpha - r, \alpha + r]$, where $r > 0$ and $f(\alpha) = 0$. Suppose $|f'(x)| \geq m > 0$ and $|f''(x)| \leq M$ on that closed interval, with $q = Mr/(2m) < 1$. For every x_0 in the interval, the exact Newton iteration $x_{k+1} = x_k - f(x_k)/f'(x_k)$ is well defined, stays in the interval, and converges to α, with $|x_{k+1} - \alpha| \leq (M/(2m))|x_k - \alpha|^2$.*

A. Foundations and permitted imports

Use the fundamental theorem of calculus and its estimate $|f'(u) - f'(v)| \leq M|u - v|$ on the interval, obtained by integrating f''. No existence of a root is inferred from numerical residuals: the theorem assumes the root α and proves convergence near it. Every simple root has some neighborhood satisfying these hypotheses by continuity and a sufficiently small radius.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. For $N(x) = x - f(x)/f'(x)$, $|N(x) - \alpha| \leq (M/(2m))|x - \alpha|^2$.

Receiving activity. N2.1

Compatibility. E1/v1 and $q = Mr/(2m) < 1$. Check: Proving the estimate at one point is insufficient unless its next iterate stays where the hypotheses hold.

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Derive the exact error estimate

Object and role. Subtract the secant integral from the tangent term and bound the derivative variation. **Local obligation:** For $N(x) = x - f(x)/f'(x)$, $|N(x) - \alpha| \le (M/(2m))|x - \alpha|^2$.

Inputs. $x \in [\alpha - r, \alpha + r]$ and $f(\alpha) = 0$.

E — Ego

Publish E1/v1 only after establishing this obligation: For $N(x) = x - f(x)/f'(x)$, $|N(x) - \alpha| \le (M/(2m))|x - \alpha|^2$.

M — Memory

$x \in [\alpha - r, \alpha + r]$ and $f(\alpha) = 0$. Retain the stated assumptions and import versions.

I — Intellect

Subtract the secant integral from the tangent term and bound the derivative variation.

S — Sense

The denominator stays away from zero by m; the entire integration segment stays in the stated interval.

A — Action

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Let $e = x - \alpha$. The fundamental theorem gives $f(x) = \int_\alpha^x f'(t)\,dt$. Therefore $N(x) - \alpha = [ef'(x) - f(x)]/f'(x) = \big(\int_\alpha^x (f'(x) - f'(t))\,dt\big)/f'(x)$.

N1.2 The absolute numerator is at most the integral of $M|x - t|$ along the segment, namely $M|e|^2/2$. Divide by $|f'(x)| \ge m$ to obtain the error estimate. This remains valid for $x < \alpha$ after taking the absolute value of the oriented integral.

Export E1/v1. For $N(x) = x - f(x)/f'(x)$, $|N(x) - \alpha| \le (M/(2m))|x - \alpha|^2$.

Acceptance gate. The denominator stays away from zero by m; the entire integration segment stays in the stated interval.

Targeted repair. If a Taylor remainder is used without a bound over the whole segment, use this integral identity with the explicit uniform bound M. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Close the invariant and convergence

Object and role. Inductively keep every error within the radius, then sum no errors at all: directly contract their magnitudes. **Local obligation:** All iterates exist, $|x_k - \alpha| \le q^k |x_0 - \alpha| \to 0$, and the quadratic bound holds at every step.

Inputs. E1/v1 and $q = Mr/(2m) < 1$. Incoming proof exports: E1/v1 at N2.1.

E — Ego

Publish E2/v1 only after establishing this obligation: All iterates exist, $|x_k - \alpha| \le q^k |x_0 - \alpha| \to 0$, and the quadratic bound holds at every step.

M — Memory

E1/v1 and $q = Mr/(2m) < 1$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

I — Intellect

Inductively keep every error within the radius, then sum no errors at all: directly contract their magnitudes.

S — Sense

Proving the estimate at one point is insufficient unless its next iterate stays where the hypotheses hold.

A — Action

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 If $|x_k - \alpha| \le r$, E1 gives $|x_{k+1} - \alpha| \le (M/(2m))|x_k - \alpha|^2 \le q|x_k - \alpha| \le qr < r$ (with immediate equality to zero allowed). The initial point satisfies the premise, so induction proves that every denominator is nonzero and every iterate remains in the interval.

N2.2 The same estimate yields $|x_k - \alpha| \le q^k |x_0 - \alpha|$, proving convergence. At each step E1 supplies the stronger quadratic error inequality. If $M = 0$, the first step reaches the root exactly. Finally a simple root has $|f'(\alpha)| > 0$; continuity gives a neighborhood with a positive derivative bound, bounded f'', and after shrinking the radius, $q < 1$.

Export E2/v1. All iterates exist, $|x_k - \alpha| \le q^k |x_0 - \alpha| \to 0$, and the quadratic bound holds at every step.

Acceptance gate. Proving the estimate at one point is insufficient unless its next iterate stays where the hypotheses hold.

Targeted repair. If local convergence is concluded while the iterates may leave the neighborhood, choose and prove an invariant radius before repeating the estimate. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Derive the exact error estimate

1. **N1.1.** Let $e = x - \alpha$. The fundamental theorem gives $f(x) = \int_\alpha^x f'(t)\,dt$. Therefore $N(x) - \alpha = [ef'(x) - f(x)]/f'(x) = \big(\int_\alpha^x (f'(x) - f'(t))\,dt\big)/f'(x)$.

2. **N1.2.** The absolute numerator is at most the integral of $M|x - t|$ along the segment, namely $M|e|^2/2$. Divide by $|f'(x)| \geq m$ to obtain the error estimate. This remains valid for $x < \alpha$ after taking the absolute value of the oriented integral.

B. N2 — Close the invariant and convergence

1. **N2.1.** If $|x_k - \alpha| \leq r$, E1 gives $|x_{k+1} - \alpha| \leq (M/(2m))|x_k - \alpha|^2 \leq q|x_k - \alpha| \leq qr < r$ (with immediate equality to zero allowed). The initial point satisfies the premise, so induction proves that every denominator is nonzero and every iterate remains in the interval.

2. **N2.2.** The same estimate yields $|x_k - \alpha| \leq q^k|x_0 - \alpha|$, proving convergence. At each step E1 supplies the stronger quadratic error inequality. If $M = 0$, the first step reaches the root exactly. Finally a simple root has $|f'(\alpha)| > 0$; continuity gives a neighborhood with a positive derivative bound, bounded f'', and after shrinking the radius, $q < 1$.

□

V. WORKED ORCHESTRATION

For $f(x) = x^2 - 2$, take $\alpha = \sqrt{2}$ and $r = 1/4$. On that interval $|f'| \geq 2(\sqrt{2} - 1/4) > 2$, so $m = 2$ is valid; $M = 2$ and $q = 1/8$. Starting at $x_0 = 3/2$ gives $x_1 = 17/12$ and $x_2 = 577/408$. The exact identity $x_{k+1} - \sqrt{2} = (x_k - \sqrt{2})^2/(2x_k)$ displays the quadratic mechanism.

A. Boundary and hypothesis audit

At the multiple root of $f(x) = x^2$, Newton gives $x_{k+1} = x_k/2$ for $x_k \neq 0$, only linear convergence. The derivative lower bound fails in every neighborhood of zero. Starting far away can also break local convergence; this theorem does not promise a global basin.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 claim: the quadratic error formula proves convergence from every initial guess. **Sense:** its constants were established only on a fixed interval, and outside it the denominator may vanish. **Repair:** prove the radius invariant using $q < 1$ before iteration. This supplies the missing step for the stated local theorem; it does not extend the theorem globally. Recheck each use of E1 along the orbit.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend the argument to vector Newton methods using an invertible Jacobian and a controlled inverse, with explicit matrix norm bounds.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Boyd and Vandenberghe, Convex Optimization for convexity, optimality, and descent; University of Maryland, Scientific Computing I for the numerical-analysis setting. The estimates used here are derived explicitly in §4.